

DE-2026 — PREDICTED QUESTION PAPER

Gujarat Technological University · Diploma Engineering Entrance

Questions: 100 | Total Marks: 200 | Time: 150 Minutes | +2 per correct | –0.5 negative

SECTION 1 — Basics of Science and Engineering (Q.01 to Q.50 · 100 Marks)

UNITS & MEASUREMENT — Q.01 TO Q.06

Q.01 What is the SI unit of electric potential?

- A. Ampere
- B. Volt
- C. Coulomb
- D. Watt

Q.02 If the percentage error in the measurement of mass is 1% and in radius is 2%, what is the maximum percentage error in the density of a sphere?

- A. 3%
- B. 5%
- C. 7%
- D. 1%

Q.03 1 Joule is equal to _____ erg.

- A. 10^5
- B. 10^{-5}
- C. 10^7
- D. 10^{-7}

Q.04 A micrometer screw has a pitch of 1 mm and 50 divisions on its circular scale. What is its least count?

- A. 0.01 mm
- B. 0.02 mm
- C. 0.001 mm
- D. 0.05 mm

Q.05 The dimensional formula for Force is:

- A. $M^1L^1T^{-1}$
- B. $M^1L^1T^{-2}$
- C. $M^1L^2T^{-2}$
- D. $M^1L^2T^{-1}$

Q.06 Which of the following is not a fundamental physical quantity?

- A. Length
- B. Time
- C. Mass
- D. Force

CLASSICAL MECHANICS — Q.07 TO Q.12

Q.07 Newton's first law of motion defines:

- A. Force
- B. Inertia
- C. Momentum
- D. Acceleration

Q.08 If a body of mass 5 kg is moving in a circular path of radius 2 m with a constant speed of 10 m/s, what is the centripetal force acting on it?

- A. 50 N
- B. 100 N
- C. 250 N
- D. 500 N

Q.09 The rate of change of linear momentum of a body is equal to:

- A. Impulse
- B. Work done
- C. External applied force
- D. Kinetic energy

Q.10 What is the kinetic energy of a body of mass 2 kg moving with a velocity of 5 m/s?

- A.** 10 J
- B.** 25 J
- C.** 50 J
- D.** 5 J

Q.11 Work done by the centripetal force on a body moving in a uniform circular motion is:

- A.** Maximum
- B.** Minimum
- C.** Zero
- D.** Infinite

Q.12 Angular velocity is measured in:

- A.** m/s
- B.** rad/s
- C.** degrees/s²
- D.** rad/s²

ELECTRIC CURRENT — Q.13 TO Q.18

Q.13 Three resistors of 6 Ω each are connected in parallel. What is the equivalent resistance?

- A.** 18 Ω
- B.** 6 Ω
- C.** 3 Ω
- D.** 2 Ω

Q.14 The electric flux passing through a surface depends on:

- A.** Electric field intensity only
- B.** Area of the surface only
- C.** Angle between electric field and surface normal
- D.** All of the above

Q.15 If the distance between two point charges is halved, the electrostatic force between them becomes:

- A.** Halved
- B.** Doubled
- C.** Four times
- D.** One-fourth

Q.16 Which device is used to store electric charge?

- A.** Resistor
- B.** Inductor
- C.** Capacitor
- D.** Galvanometer

Q.17 According to Ohm's Law, the graph between V and I for a metallic conductor is a:

- A.** Parabola
- B.** Straight line passing through origin
- C.** Circle
- D.** Hyperbola

Q.18 The unit of electrical resistivity is:

- A.** Ohm
- B.** Ohm/meter
- C.** Ohm-meter
- D.** Mho

HEAT & THERMOMETRY — Q.19 TO Q.24

Q.19 The normal temperature of the human body is 98.6°F. What is it in Celsius?

- A.** 37°C
- B.** 36°C
- C.** 38°C
- D.** 39°C

Q.20 Heat transfer by conduction takes place primarily in:

- A.** Liquids
- B.** Gases
- C.** Solids
- D.** Vacuum

Q.21 The specific heat of water in SI units is approximately:

- A.** 4.186 J/kg K
- B.** 4186 J/kg K
- C.** 41.86 J/kg K
- D.** 1 J/kg K

Q.22 Absolute zero temperature is equivalent to:

- A.** 0°C
- B.** 273.15°C
- C.** -273.15°C
- D.** -100°C

Q.23 Which of the following has the highest thermal conductivity?

- A.** Copper
- B.** Aluminum
- C.** Iron
- D.** Silver

Q.24 The ratio of the amount of heat supplied to a body to the change in its temperature is called:

- A.** Specific heat
- B.** Heat capacity
- C.** Latent heat
- D.** Thermal conductivity

WAVE MOTION, OPTICS & ACOUSTICS — Q.25 TO Q.30

Q.25 Sound waves in air are:

- A.** Transverse mechanical waves
- B.** Longitudinal mechanical waves
- C.** Electromagnetic waves
- D.** Non-mechanical waves

Q.26 The reverberation time of a hall with a volume of 3000 m³ and a total sound absorption of 600 O.W.U. is approximately:

- A.** 0.805 sec
- B.** 1.205 sec
- C.** 5.000 sec
- D.** 0.500 sec

Q.27 Total internal reflection can occur only when light travels from:

- A.** Rarer to denser medium
- B.** Denser to rarer medium
- C.** One medium to another of same refractive index
- D.** Vacuum to air

Q.28 Which of the following is a characteristic of LASER light?

- A.** Monochromatic
- B.** Coherent
- C.** Unidirectional
- D.** All of the above

Q.29 Optical fibers are widely used in telecommunications. They work on the principle of:

- A.** Dispersion of light
- B.** Refraction of light
- C.** Total internal reflection
- D.** Polarization of light

Q.30 If the frequency of a sound wave is 500 Hz and its speed is 340 m/s, what is its wavelength?

- A.** 1.47 m
- B.** 0.68 m
- C.** 680 m
- D.** 170000 m

CHEMICAL REACTIONS & EQUATIONS — Q.31 TO Q.33

Q.31 Identify the balanced chemical equation for the reaction of iron with steam:

- | | |
|--|--|
| A. $\text{Fe} + \text{H}_2\text{O} \rightarrow \text{FeO} + \text{H}_2$ | B. $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$ |
| C. $2\text{Fe} + 3\text{H}_2\text{O} \rightarrow \text{Fe}_2\text{O}_3 + 3\text{H}_2$ | D. $3\text{Fe} + 2\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 2\text{H}_2$ |

Q.32 The reaction $\text{Pb}(\text{NO}_3)_2(\text{aq}) + 2\text{KI}(\text{aq}) \rightarrow \text{PbI}_2(\text{s}) + 2\text{KNO}_3(\text{aq})$ is an example of:

- | | |
|--|----------------------------------|
| A. Combination reaction | B. Decomposition reaction |
| C. Double displacement reaction | D. Oxidation reaction |

Q.33 Digestion of food in the human body is an example of which type of reaction?

- | | |
|------------------------|-------------------------------|
| A. Displacement | B. Decomposition |
| C. Combination | D. Double displacement |

ACIDS, BASES & SALTS — Q.34 TO Q.37

Q.34 What is the pH value of a neutral solution like pure water at 25°C?

- | | |
|--------------|-------------|
| A. 0 | B. 7 |
| C. 14 | D. 1 |

Q.35 The chemical formula for Bleaching Powder is:

- | | |
|---|----------------------------|
| A. $\text{CaSO}_4 \cdot (1/2)\text{H}_2\text{O}$ | B. CaCO_3 |
| C. CaOCl_2 | D. NaHCO_3 |

Q.36 When an acid reacts with a metal, which gas is typically liberated?

- | | |
|--------------------|--------------------------|
| A. Oxygen | B. Carbon dioxide |
| C. Hydrogen | D. Nitrogen |

Q.37 Milk of Magnesia is often used as an antacid. Its chemical name is:

- | | |
|-------------------------------|-------------------------------|
| A. Magnesium oxide | B. Magnesium hydroxide |
| C. Magnesium carbonate | D. Magnesium sulfate |

METALS & NON-METALS — Q.38 TO Q.40

Q.38 Which of the following metals is the most reactive?

- | | |
|----------------|---------------------|
| A. Gold | B. Iron |
| C. Zinc | D. Potassium |

Q.39 Which non-metal is a liquid at room temperature?

- | | |
|--------------------|-------------------|
| A. Iodine | B. Bromine |
| C. Chlorine | D. Carbon |

Q.40 The process of heating a sulphide ore in the presence of excess air to convert it into an oxide is known as:

- | | |
|-----------------------|--------------------|
| A. Calcination | B. Roasting |
| C. Reduction | D. Smelting |

COMPUTER PRACTICE — Q.41 TO Q.45

Q.41 In MS-Word, the keyboard shortcut to Undo an action is:

- | | |
|--------------------|--------------------|
| A. Ctrl + Y | B. Ctrl + Z |
| C. Ctrl + X | D. Ctrl + C |

Q.42 1 Gigabyte (GB) is equal to:

- | | |
|-------------------------------|-------------------------------|
| A. 1024 Megabytes (MB) | B. 1000 Megabytes (MB) |
| C. 1024 Kilobytes (KB) | D. 1024 Bytes |

Q.43 The correct HTML tag to insert a line break is:

- | | |
|----------------|-------------------|
| A. [lb] | B. [break] |
| C. [br] | D. [tr] |

Q.44 In MS-Excel, formulas must always begin with which symbol?

- | | |
|-------------|-------------|
| A. + | B. - |
| C. = | D. @ |

Q.45 Which part of the computer is considered its brain?

- | | |
|---------------|----------------------|
| A. RAM | B. ROM |
| C. CPU | D. Hard Drive |

ENVIRONMENTAL SCIENCES — Q.46 TO Q.50

Q.46 Which gas is most responsible for the greenhouse effect and global climate change?

- | | |
|--------------------------|--------------------|
| A. Oxygen | B. Nitrogen |
| C. Carbon dioxide | D. Argon |

Q.47 The ozone layer, which protects the earth from harmful UV radiation, is found in which atmospheric layer?

- | | |
|-----------------------|------------------------|
| A. Troposphere | B. Stratosphere |
| C. Mesosphere | D. Thermosphere |

Q.48 Which of the following is a non-renewable source of energy?

- | | |
|-----------------------------|-----------------------|
| A. Solar Energy | B. Wind Energy |
| C. Geothermal Energy | D. Coal |

Q.49 Acid rain is primarily caused by the emissions of which of the following gases?

- | | |
|--|--|
| A. Carbon dioxide and Carbon monoxide | B. Sulfur dioxide and Nitrogen oxides |
| C. Methane and Ozone | D. CFCs |

Q.50 The energy derived from the internal heat of the earth is known as:

- | | |
|-----------------------------|------------------------|
| A. Biomass energy | B. Tidal energy |
| C. Geothermal energy | D. Hydro energy |

SECTION 2 — Aptitude Test: Mathematics + Soft Skills (Q.51 to Q.100 · 100 Marks)

DETERMINANTS & MATRICES — Q.51 TO Q.54

Q.51 If matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and matrix $B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$, what is the element in the first row and first column of AB ?

- A.** 2
- B.** 4
- C.** 5
- D.** 7

Q.52 The determinant of an identity matrix of order 3×3 is:

- A.** 0
- B.** 1
- C.** 3
- D.** 9

Q.53 A matrix is said to be a singular matrix if its determinant is:

- A.** Equal to 1
- B.** Less than 0
- C.** Equal to 0
- D.** Greater than 1

Q.54 If A is a 2×3 matrix and B is a 3×4 matrix, what is the order of the product matrix AB ?

- A.** 3×3
- B.** 2×4
- C.** 4×2
- D.** Matrix multiplication is not possible

TRIGONOMETRY — Q.55 TO Q.57

Q.55 Convert $5\pi/6$ radians to degrees.

- A.** 120°
- B.** 150°
- C.** 210°
- D.** 300°

Q.56 The value of $\sin(90^\circ + \theta)$ is equal to:

- A.** $\sin \theta$
- B.** $-\sin \theta$
- C.** $\cos \theta$
- D.** $-\cos \theta$

Q.57 The principal period of the function $f(x) = \tan(2x)$ is:

- A.** π
- B.** $\pi/2$
- C.** 2π
- D.** $\pi/4$

VECTORS — Q.58 TO Q.59

Q.58 If the dot product of two non-zero vectors is zero, then the angle between them is:

- A.** 0°
- B.** 45°
- C.** 90°
- D.** 180°

Q.59 The magnitude of the vector $3\mathbf{i} - 4\mathbf{j} + 12\mathbf{k}$ is:

- A.** 11
- B.** 13
- C.** 17
- D.** 19

COORDINATE GEOMETRY — Q.60 TO Q.61

Q.60 The distance between the points $(2, 3)$ and $(-1, -1)$ is:

- A.** 3
- B.** 4
- C.** 5
- D.** 7

Q.61 The center of the circle represented by the equation $x^2 + y^2 + 6x - 8y + 9 = 0$ is:

- A. (3, -4)
- B. (-3, 4)
- C. (6, -8)
- D. (-6, 8)

FUNCTION & LIMIT — Q.62 TO Q.64

Q.62 Evaluate: $\lim_{x \rightarrow 0} (\tan 5x) / x$

- A. 1
- B. 5
- C. $1/5$
- D. 0

Q.63 A function $f(x)$ is called an even function if:

- A. $f(-x) = -f(x)$
- B. $f(-x) = f(x)$
- C. $f(x) = 1/f(x)$
- D. $f(-x) = 0$

Q.64 Evaluate: $\lim_{x \rightarrow 3} (x^2 - 9) / (x - 3)$

- A. 0
- B. 3
- C. 6
- D. Limit does not exist

DIFFERENTIATION & APPLICATIONS — Q.65 TO Q.68

Q.65 If $y = e^{(3x)}$, then dy/dx is:

- A. $e^{(3x)}$
- B. $3e^{(3x)}$
- C. $3x * e^{(3x)}$
- D. $e^{(3x)}/3$

Q.66 The derivative of $\log(\sin x)$ with respect to x is:

- A. $1/\sin x$
- B. $\cos x$
- C. $\tan x$
- D. $\cot x$

Q.67 If $x = t^2$ and $y = 2t$, then dy/dx is:

- A. t
- B. $1/t$
- C. $2/t$
- D. $2t$

Q.68 The second derivative of $f(x) = x^4$ with respect to x is:

- A. $4x^3$
- B. $12x^2$
- C. $24x$
- D. $x^3/3$

INTEGRATION — Q.69 TO Q.72

Q.69 Evaluate the integral: integral of $x^3 dx$

- A. $3x^2 + C$
- B. $(x^4/4) + C$
- C. $(x^4/3) + C$
- D. $x^4 + C$

Q.70 Evaluate the integral: integral of $e^{(2x)} dx$

- A. $e^{(2x)} + C$
- B. $2e^{(2x)} + C$
- C. $(e^{(2x)})/2 + C$
- D. $e^x + C$

Q.71 Evaluate the definite integral: integral from 0 to 1 of $2x dx$

- A. 1
- B. 2
- C. 0
- D. $1/2$

Q.72 Evaluate the integral: integral of $(1/x) dx$

- A.** $x^{(-2)} + C$
- B.** $-1/x^2 + C$
- C.** $\log(x) + C$
- D.** $e^x + C$

LOGARITHM — Q.73 TO Q.74

Q.73 If $\log_2(x) = 5$, then the value of x is:

- A.** 10
- B.** 25
- C.** 32
- D.** 64

Q.74 The value of $\log(10) + \log(100)$ is:

- A.** 1000
- B.** 110
- C.** 3
- D.** 2

STATISTICS — Q.75

Q.75 The mean of the first 5 prime numbers is:

- A.** 3.6
- B.** 5.6
- C.** 4.2
- D.** 6.4

COMPREHENSION OF UNSEEN PASSAGE — Q.76 TO Q.80

Read the following passage carefully and answer Q.76 to Q.80: The industrial revolution, which began in Britain in the late 18th century, marked a major turning point in history. It shifted societies from agrarian and handicraft-based economies to ones dominated by industry and machine manufacturing. The invention of the steam engine by James Watt was pivotal, powering factories and revolutionizing transportation through steamships and railways. While it brought unprecedented economic growth and technological advancement, it also led to severe urbanization problems, poor working conditions, and environmental pollution. The soot-covered cities became a stark contrast to the green countryside, fundamentally altering the human experience and paving the way for the modern world.

Q.76 Where did the industrial revolution begin?

- A.** France
- B.** Germany
- C.** Britain
- D.** United States

Q.77 Which invention was pivotal for powering factories during this era?

- A.** The printing press
- B.** The cotton gin
- C.** The steam engine
- D.** The telegraph

Q.78 The industrial revolution shifted societies away from what type of economies?

- A.** Information-based economies
- B.** Agrarian and handicraft-based economies
- C.** Globalized economies
- D.** Service-oriented economies

Q.79 According to the passage, what was one negative consequence of the industrial revolution?

- A.** Unprecedented economic growth
- B.** Decrease in population
- C.** Environmental pollution
- D.** A return to agrarian lifestyles

Q.80 The passage states that the soot-covered cities became a stark contrast to:

- A.** The green countryside
- B.** The bustling ports
- C.** The modern skyscrapers
- D.** The clean factories

THEORY OF COMMUNICATION — Q.81 TO Q.85

Q.81 In the communication process, the receiver's response to the sender's message is known as:

- A.** Encoding
- B.** Channeling
- C.** Noise
- D.** Feedback

Q.82 The study of spatial distance between individuals during communication is termed:

- A.** Kinesics
- B.** Proxemics
- C.** Chronemics
- D.** Haptics

Q.83 Which of the following is considered a psychological barrier to communication?

- A.** Loud background noise
- B.** Faulty translation
- C.** Prejudice and emotions
- D.** Defective telephone line

Q.84 Gestures, eye contact, and posture are all examples of:

- A.** Verbal communication
- B.** Written communication
- C.** Non-verbal communication
- D.** Semantic communication

Q.85 The process of converting an idea into a structured message by the sender is called:

- A.** Decoding
- B.** Encoding
- C.** Transmitting
- D.** Receiving

TECHNIQUES OF WRITING — Q.86 TO Q.90

Q.86 In a formal business letter, where is the 'Inside Address' located?

- A.** Below the signature
- B.** At the top right corner
- C.** Above the salutation on the left
- D.** Within the body paragraph

Q.87 Which of the following is an appropriate complimentary close for a formal business letter starting with 'Dear Sir/Madam'?

- A.** Yours lovingly,
- B.** Yours faithfully,
- C.** Best wishes,
- D.** Cheers,

Q.88 If a document is attached along with a business letter, it is indicated at the bottom by the abbreviation:

- A.** P.S.
- B.** C.C.
- C.** Encl.
- D.** N.B.

Q.89 An 'Order Letter' must contain accurate details regarding:

- A.** The history of the company
- B.** Quantity, catalog number, and price of goods
- C.** The personal life of the manager
- D.** General inquiries about future products

Q.90 What does C.C. stand for in email and letter formatting?

- A.** Correct Copy
- B.** Carbon Copy
- C.** Checked Copy
- D.** Courtesy Communication

GRAMMAR — Q.91 TO Q.95

Q.91 The committee _____ decided to postpone the meeting until next week.

- A.** have
- B.** has
- C.** are
- D.** were

Q.92 She has been working on this project _____ Monday.

- A.** from
- B.** for
- C.** since
- D.** untill

Q.93 "Alas! We have lost the match." Identify the interjection.

- A.** We
- B.** match
- C.** lost
- D.** Alas!

Q.94 He speaks English _____. (Select the correct adverb)

- A.** fluent
- B.** fluency
- C.** fluently
- D.** more fluent

Q.95 Neither the manager nor the employees _____ aware of the new policy.

- A.** is
- B.** was
- C.** are
- D.** has

CORRECTION OF WORDS & SENTENCES — Q.96 TO Q.100

Q.96 Select the correct spelling.

- A.** Accomodation
- B.** Accommodation
- C.** Acommodation
- D.** Acomodation

Q.97 Select the correct spelling.

- A.** Maintanance
- B.** Maintenance
- C.** Maintenence
- D.** Maintainance

Q.98 Identify the grammatically correct sentence.

- A.** He is one of the best student in the class.
- B.** He is one of the best students in the class.
- C.** He are one of the best student in the class.
- D.** He is one best students in the class.

Q.99 Identify the grammatically correct sentence.

- A.** I prefer coffee than tea.
- B.** I prefer coffee over tea.
- C.** I prefer coffee to tea.
- D.** I prefer coffee from tea.

Q.100 Identify the grammatically correct sentence.

- A.** If I was a bird, I would fly.
- B.** If I am a bird, I will fly.
- C.** If I were a bird, I would fly.
- D.** If I had been a bird, I fly.